

Genetic Algorithm Based Maximum Power Tracking in Solar Power Generation

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Abstract— This paper presents genetic algorithm (GA) based maximum power point tracking (MPPT) for photovoltaic (PV) array integrated with battery storage unit (BSU) as power generation unit in standalone mode. PV generation depends on solar irradiance, site location and environmental factors like temperature, wind and. Thus output of PV output is fluctuating in nature and the addition of non linear load make the situation more critical. GA based MPPT for PV generation works for local optimal solution. DC/DC and boost converter has been used to obtain rated desired rated voltage. BSU acts as secondary source to ensure uninterrupted power supply. A suitable three phase voltage source inverter (VSI), at point of common coupling (PCC), provides regulated power supply. Boost converters incorporating proportional controller (PI) increases PV voltage up to 400V. Conventionally used perturb and observe (PO) algorithm has been used for comparative study. Controlled pulse width modulated (PWM) switching pulse for converter unit and single section LC filter at output terminal eliminates harmonic components. To exhibit the system compatibility resistive, reactive, asynchronous induction motor and nonlinear load has been switched at different instants for variable solar irradiance and temperature.

Index Terms— battery storage, fuel cell, MPPT, genetic algorithm, PV generation, voltage source converter.

I. INTRODUCTION

The high solar potential, i.e. about 20,000 times the total world demand of electricity, has not yet been exploited due to technical and economical lags. Solar energy is omnipresent and available round the globe free of cost. Beside all these advantages, PV generation is sure to remain unavailable during night time. Therefore, secondary power source like BSU need to be integrated with PV to assure uninterrupted power supply in standalone mode. BSU also acts as power conditioner and integrated VSI minimizes THD as well as works as active power filter. In order to track maximum power a number of approaches, commonly known as maximum power point tracker (MPPT), have been made. Among the commonly applied MPPT approaches; like perturb & observe (PO), power matching, incremental conductance, fractional open-circuit voltage/short-circuit current, power differential feedback control, curve fitting, dc-link capacitor droop control, intelligent control, and some other special control methods; PO is commonly used due to its simplicity and good results [1-9].

Due to dynamic nature of GA, it has drawn attention of power system researchers to explore and exploit renewable energy sources optimally. This paper deals with comparative

study of GA and PO for exploration and exploitation of renewable energy sources optimally.

II. MODEL OF HYBRID PV-BSU SYSTEM

Dynamic PV hybrid model consisting of individual PV array and BSU is connected to point of common coupling (PCC) through bus bars. It is assumed that PV output power changes round the clock. Partial shading condition has been neglected, i.e. PV array is either fully exposed to sun light or fully shaded, to avoid reverse biasing condition or local heat spot formation in PV module [10]. The modern PV module consists of PV cells, mounting structure, MPPT mechanism, converters, storage unit, electrical & mechanical connections to regulate and utilize electrical output [11]. The electrical power output of PV module depends on electrical, thermal, PV spectral and optical property of PV cell array and PV angle & irradiance [12]. The technical problems such as islanding detection, harmonic distortion, electromagnetic interference and low efficiency of PV cells are major bottle neck for widespread application of PV systems [13].

A. PV Power Generation

Conventionally PV cells were designed based on band gap energy (eV). Low band gap energy have high current ($I=eNA$) but low voltage ($V=E_g/e$), and vice versa. Here, e is electron charge, N is the number of photons, E_g is energy gap, and A is the surface area of solar cell. Therefore, it was preferred to use materials with energy gap between 1 and 1.8 eV like crystal silicon (1.12 eV), amorphous silicon (1.75 eV), copper indium diselenide (1.05eV), cadmium telluride (1.45 eV), gallium arsenide (1.42 eV) and indium phosphide (1.34 eV) [14, 15]. These modules are rated in terms of peak kilowatts (KW_p) i.e., the amount of expected electrical power output when the sun is directly overhead on a clear day.

A solar cell is generally represented by one diode model as shown in fig. 1.

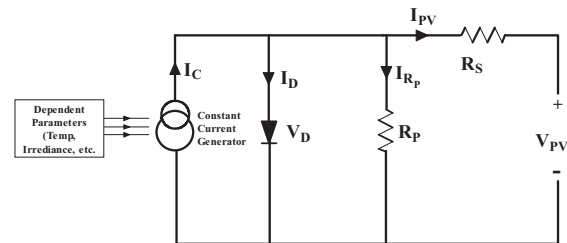


Fig: 1 One diode model of PV cell

The model includes a constant current source representing PV cell, a diode, a shunt resistance R_p and a series resistance R_s . The basic terms involved in PV conversion are given as:

$$I_{PV} = I_C - I_D - \frac{V_D}{R_p} = I_C - I_D - I_{R_p} \quad (1)$$

$$V_{PV} = V_D - R_s I_{PV} \quad (2)$$

$$I_D = I_0 \left(e^{(V_D/nkT)} - 1 \right) \quad (3)$$

$$T = T_a + \left(\frac{NOCT - 20^0}{0.8} \right) S_i \quad (4)$$

Where, I_0 is saturation current, V_D is diode voltage, I_D is diode current, k is Boltzmann's constant ($1.38 \times 10^{-23} \text{ J/K}$), n is numerical constant (1 for germanium and 2 for silicon), T is junction temperature, T_a is ambient temperature, and $NOCT$ is cell temperature in a module when ambient temperature is 20^0 C .

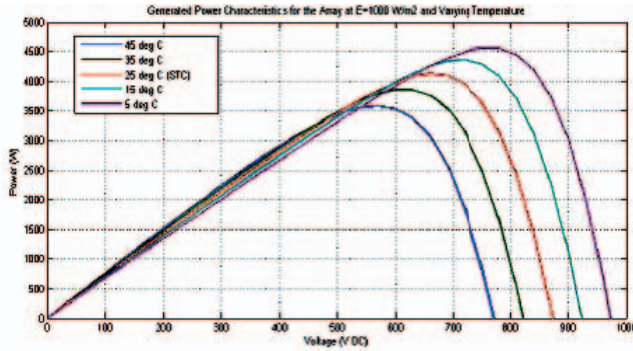


Fig:2 Change in the characteristics due to the change in temperature

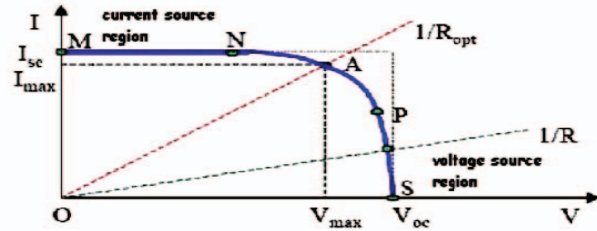


Fig: 3 Typical I-V characteristic of a solar cell in steady-state operation

Conversion efficiency (η) of PV cell is given as the ratio of maximum power output (P_{mo}) and input power (P_{in}) at a given cell temperature T .

$$\eta = \frac{P_{mo}}{P_{in}} = \frac{P_{mo}}{AS_i} = \frac{(V_{oc} \cdot I_{sc}) F_F}{AS_i} \quad (5)$$

$$\text{Where, } V_{oc} = \frac{nkT}{e} \left[\log \left(\frac{I_L + I_0}{I_0} \right) \right] \quad (6)$$

$$I_{sc} = I_{PV} - I_0 \left[e^{(eR_s I_{sc}/nkT)} - 1 \right] - \left(\frac{R_s I_{sc}}{R_{sh}} \right) \quad (7)$$

$$\text{And, } F_F = \frac{V_m \cdot I_m}{V_{oc} \cdot I_{sc}} = \frac{P_m}{V_{oc} \cdot I_{sc}} \quad (8)$$

Here, S_i is solar irradiance, V_{oc} is open circuit voltage, V_{sc} is short circuit voltage, and F_F is fill factor (0.5 to 1).

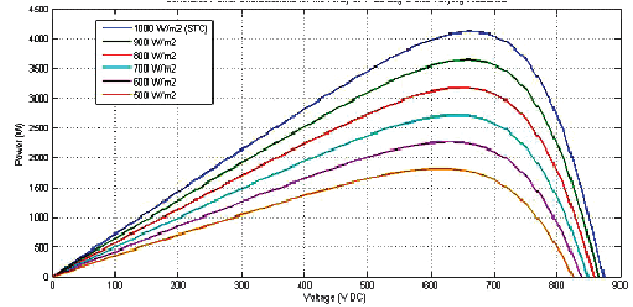


Fig: 4 Change in the characteristics due to the change in solar insolation

B. Battery Storage Unit

Due to simplicity and robust working if lead acid battery; it is generally used with PV module. The equivalent circuit of the battery is shown in fig 5. The parameters of battery have been given in appendix.

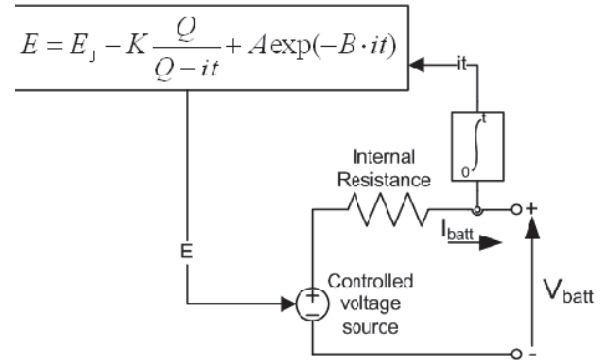


Fig. 5 Equivalent circuit of the battery

III. OBJECTIVE FUNCTION FORMULATION

The constraints taken under consideration include power of the PV module and battery. GA and PO aim at tracking maximum power output and minimizing the objective function to global optimal result. Algorithm traces global maxima and minima by voltage versus power curve. Power balance constraints for any period includes power supplied by the hybrid generation and load demand, i.e. generated power must meet total load demand, satisfying power quality and reliability constraints.

$$P_{PV} + P_{BSU} \geq P_{load} \quad (9)$$

From power curve of selected PV module, it is obtained that voltage increases from 0 to V_{pv} . Power curve first increases to a maximum point and then gradually decreases to minimum value. PV module operating condition is selected so as to assure maximum power operating point. Partial shading condition has been neglected to overcome malfunctioning conditions of PV array.

In order to track MPPT with ABC, gbest-ABC, and PO; transformation of Y-axis has been suggested. Axis transformation is done so as to obtain mirror image of voltage power graph. It is given as:

$$P_{new} = (P_{PV,max} + 50) - P_{inst} \quad (10)$$

$$\text{And, } N_{PV}, N_{BSU} \geq 0 \quad (11)$$

Here P_{new} is new transformed value, $P_{pv\ max}$ is maximum power point in power curve, and P_{inst} is instantaneous power. The factor 50 is taken to avoid interference in transformation at $P_{pv,max}$ and obtain effective transformation.

For BSU, it is assured that

$$P_{BSU,min} \geq P_{BSU} \geq P_{BSU,max} \quad (12)$$

$$\text{And, } P_{BSU,min} = (1 - DOD) \cdot P_{BSU,max} \quad (13)$$

Here, $P_{BSU,min}$, P_{BSU} , and $P_{BSU,max}$ are minimum, actual and maximum capacity of BSU respectively; and DOD is depth of discharge of BSU.

IV. GENETIC ALGORITHM

A. Gbest-ABC Algorithm

GBest ABC is based on GA problem based on behavior of honey bee. It incorporates honey bees into employed and unemployed bees (UB). Number of employed bees (EB) is equal to UB. 10 to 15% of UB are scout bees (SB) and rests are onlooker bees (OB). SB search new and abandoned food source (FS) randomly without prior knowledge and gather the information about the quality of FS. As soon as SB finds new FS, it becomes EB and carries nectar to hive. It deposits nectar at hive and starts waggle dance to motivate OB. Waggle dance is one of the important activities because updated food information is shared in the dancing area. OB stays in the hive, watches waggle dance and updates food quality information. Food quality generally depends on distance of FS from hive, amount of nectar and food quantity. OB always have option of either becoming SB, stay at hive to watch waggle dance and update information gathered by EB or become EB by motivation of one of the best options among many dancing bees. When EB return to hive with nectar, they also have choice to remain an employed bee, waggle dance and motivate OB to exploit FS or become OB. This process continues until food source is depleted and all employed bees become unemployed.

Assumptions made:

1. Food quality depends on quality & quantity of nectar and distance of FS.

2. Once a bee becomes an EB, it remains EB until FS is depleted temporarily or permanently.
3. UB always look for FS by watching waggle dance, employ themselves by motivation or become SB.
4. Each FS has dedicated EB, SB and common OB.
5. EB always updates information with OB and SB regularly investigates abandoned FS.

Algorithm starts with no previous information about best FS. First of all variables like total population (TP) of bee, maximum number of cycles (MNC), objective function, and function variables are defined. Function for calculation of fitness of objective function for each FS is initiated by minimizing the greedy selection function. Depending on information of quality of nectar updated by EB and hive storage, nectar is carried by EB to the hive. Excess nectar is stored separately for future usages.

Food position gets updated in every cycle as per fitness solution by EB and SB. For known and abandoned FS, position of each FS is represented in three dimensions viz. population ($i = 1, 2, \dots, TP$), dimensional variable ($j = 1, 2, \dots, D$), and specific objective function ($k = 1, 2, \dots, OF$). Number of cycle represents number of iterations and needs to be minimized for optimal solution in a given position.

If the upper and lower bounds of i^{th} bee x_i for k^{th} variable in j^{th} direction is x_{minjk} and x_{maxjk} respectively and $\text{rand}[0, 1]$ is a uniformly distributed random number in the range $[0, 1]$; SB initiates search of new FS as:

$$X_{ijk} = x_{minjk} + \text{rand}[0,1](x_{maxjk} - x_{minjk}) \quad (14)$$

When SB finds a food source, it becomes EB and returns to hive with nectar and updates food position x_{ijk} with OB. In the consequent cycles, OB checks new position with previous position x_{pjk} and updates the information as:

$$V_{ijk} = x_{ijk} + \phi_{ijk}(x_{ijk} - x_{pjk}) \quad (15)$$

Where, i, j are randomly chosen and $i \neq j$. ϕ_{ijk} is a random number between $[-1, 1]$. After generating new position, optimal position is updated and old one is removed from memory. The probability p_i that the new FS position is optimal depends on solution fitness solution given by:

$$p_i = \frac{(\text{fitness})_i}{\sum_{n=1}^{TP} (\text{fitness})_n} \quad (16)$$

In order to achieve better feasible solution, FS must be explored and exploited optimally. ABC algorithm is good at exploration but not in terms of exploration as the probability of getting a good FS and a bad FS is equal. So by getting inspired by other algorithms like particle swarm optimization, exploitation capability of gbest ABC can be improved as:

$$V_{ijk} = x_{ijk} + \phi_{ijk}(x_{ij} - x_{pjk}) + \psi_{ijk}(y_{jk} - x_{ijk}) \quad (17)$$

Where, Ψ_{ijk} is a non-negative random number in range $[0, C]$, y_{jk} is j^{th} term of global best solution. Parameter “C” controls the rate of exploitation and exploration. Higher value of “C” leads to better exploration and lower exploitation; hence need to be taken carefully.

B. Perturb and observe Algorithm

In this algorithm a slight perturbation is introduced in the system. Due to this perturbation, module power changes. If the power increases due to the perturbation then the perturbation is continued in that direction. After the peak power is reached the power at the next instant decreases and hence after that the perturbation reverses.

When the steady state is reached the algorithm oscillates around the peak point. In order to keep the power variation small the perturbation size is kept very small. The algorithm is developed in such a manner that it sets a reference voltage of the module corresponding to the peak voltage of the module. A PI controller then acts moving the operating point of the module to that particular voltage level. It is observed that some power loss occur due to this perturbation. Also the method has a disadvantage to track the power under fast varying atmospheric conditions. Yet this algorithm is popular and considered simple.

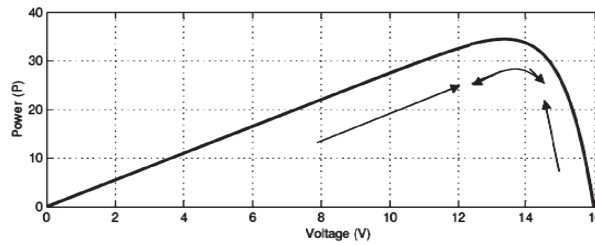


Fig. 6 Perturb and observe algorithm

V. SIMULATION RESULTS

Simulation has been done in matlab simulink platform to obtain results of hybrid PV-BSU generation system. The simulation results for PV array, MPPT buck converter, boost converter, 3-phase inverter and LC filter has been obtained.

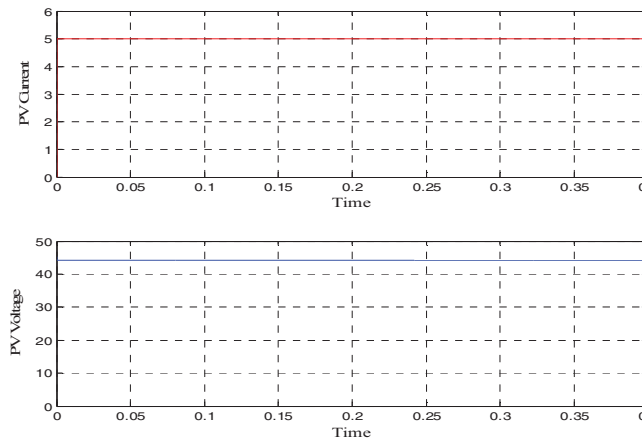


Fig. 7 PV array current and voltage for PVA alone

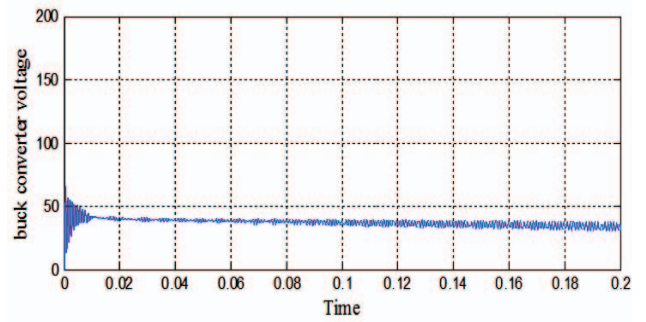


Fig. 8 MPPT buck converter output voltage

The PV array output current & voltage are shown in fig. 7. The PV array Current is 5A and voltage is 45 volts. MPPT buck converter result has been shown in fig. 8. It is observed that the MPPT buck converter maintains the PV array output to 45 volts. The output of boost converter is depicted in fig 9. The voltage is boosted to 400 volts.

BSU is used as back-up or storage purpose. Discharging characteristic is shown in fig 12. BSU used here contains lead-acid battery due to long life and simplicity. The DC link voltage is shown in fig. 11. It is also around 400 volts.

Output line voltage and current waveforms has been shown in fig 10. In fig 10, it is observed that the designed filter works satisfactorily and the output wave forms are sinusoidal.

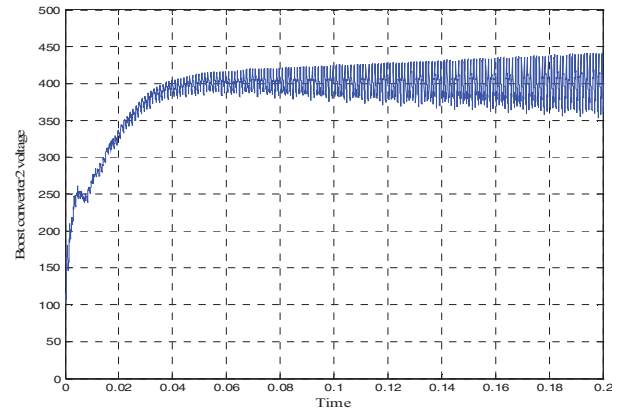


Fig. 9 Boost converter output voltage

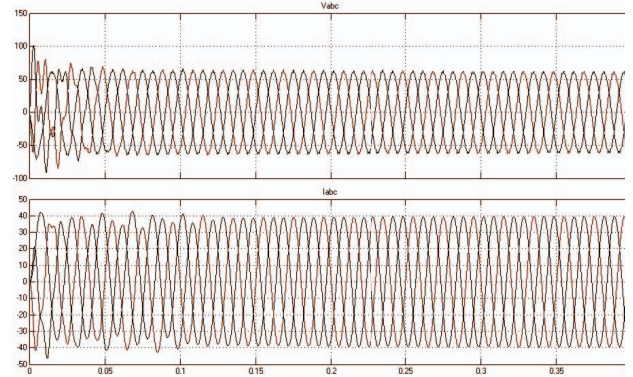


Fig. 10 Line voltage and currents wave forms

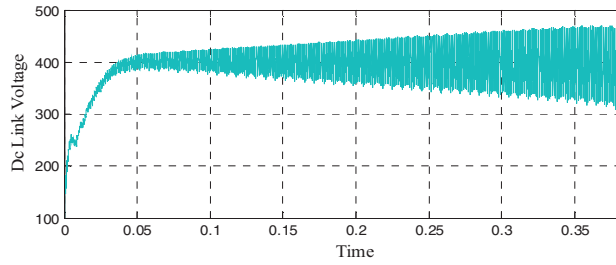


Fig. 11 DC link voltage

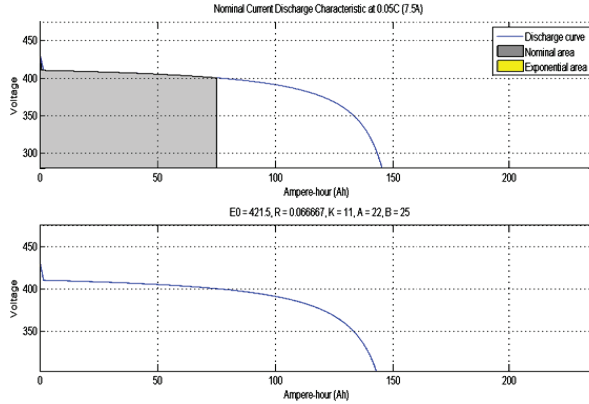


Fig. 12 Battery discharge characteristic

THD obtained for all three line voltage waveforms has been given in table 1. THD is lowest for the inductor ' $L=1.63$ mH' and capacitance ' $C=450\mu F$ '. The lowest THD is for line voltage AB (V_{ABL}) 11.88%, line voltage BC (V_{BCL}) 11.19% and line voltage CA (V_{CAL}), 5.04%

Table 1: THD at different value of L and C

S.No.	L (mH)	C (μF)	THD (V_{ABL})	THD (V_{BCL})	THD (V_{CAL})
1	1.157	250	12.31%	11.66%	5.54%
2	1.157	450	12.26%	12.01%	5.48%
3	1.63	220	12.07%	11.40%	5.38%
4	1.63	300	12.01%	11.65%	5.38%
5	1.63	250	12%	11.46%	5.31%
6	1.63	280	11.87%	11.15%	5.19%
7	1.63	330	11.77%	11.01%	5.18%
8	1.63	450	11.88%	11.19%	5.04%

Table 2: Fundamental value of line voltages at different value of L and C for PVA alone

S.No.	L (mH)	C (μF)	(V_{ABL}) rms	(V_{BCL}) rms	(V_{CAL}) rms
1	1.157	250	253.1	251.6	238.1
2	1.157	450	258.7	257.6	243.5
3	1.63	220	247.4	246.8	233
4	1.63	300	250.5	249.9	236
5	1.63	250	248.6	248	234.1
6	1.63	280	249.8	248.9	235
7	1.63	330	251.8	250.9	236.9
8	1.63	450	256.5	255.4	241.2

Magnitude of three line voltages for different values of L and C has been given in table 2. From table it is seen that fundamental values of three line voltages are maximum at $L=1.63$ mH, and $C=450\mu F$.

Fig. 13 shows comparative curve for MPPT techniques using ABC, gbest ABC, and PO using power curve for PV irradiance. PO is the most robust and simple in application but lacks exploitation of PV power at higher irradiance. At normal irradiance below 650 W/m^2 , PO has been found to exhibit better results while for higher irradiance, gbest ABC gives higher power output. Table 3 shows power obtained at different irradiance value.

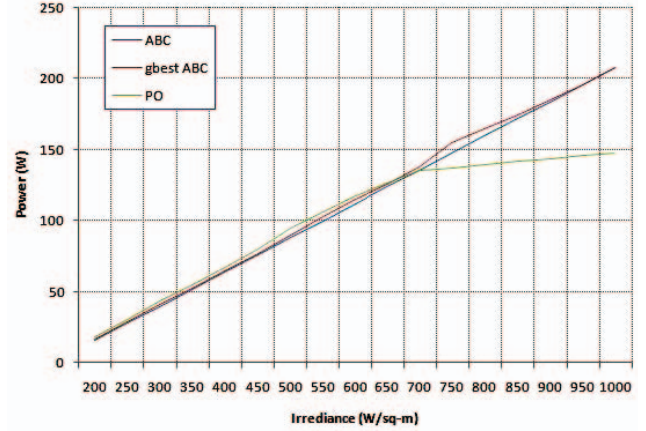


Fig. 13 Comparison of MPPT technique for ABC, gbest ABC, and PO

Table 3: Output power of PV array at different irradiance:

Irradiance	Power by different algorithms (W)		
	ABC	Gbest ABC	PO
950	196.002	196.012	146.201
750	148.879	155.013	137.658
550	100.786	103.012	106.1
250	28.346	29.048	30.003

VI. CONCLUSION

In this paper hybrid PV-BSU generation system has been studied and useful results have been accumulated. MPPT techniques based on GA has been compared and found competitive with conventional PO algorithm. GA based MPPT for PV generation works for local optimal solution. DC/DC and boost converter has been implemented successfully to obtain line voltage of 400V. Single section LC filter at output terminal eliminates harmonic components.

APPENDIX

A. Photovoltaic Array specifications

Electron charge= 1.602×10^{-19} (C)

Boltzmann constant= 1.38×10^{-23} J/oK.

Photocurrent=5A

Reverse saturation current of diode=.0002A

Series resistance of cell=.001 Ω
Reference cell operating temperature=200C

B. Battery specification

B= Exponential capacity (Ah)⁻¹ = 25
R=Internal resistance (ohms) =.066667
E0 = Constant voltage (V) =421.5
Q = Battery capacity (Ah) =150
A = Exponential voltage (V) =22

C. Boost converter2 specification

IGBT switch
Capacitor=7500 μ F
Inductor =5 μ H
Resistance=.2ohm

D. MPPT Buck converter specification

Capacitor =4400 μ F
Inductor =4 μ H
MOSFET switch

REFERENCES

- [1] S.K. Kollimalla, M.K. Mishra, "A Novel Adaptive P&O MPPT Algorithm Considering Sudden Changes in the Irradiance", *IEEE Transactions on Energy Conversion*, vol. 29, no. 3, pp. 602-610, Sept. 2014.
- [2] C. Cabal, L. Martínez-Salamero, L. Séguier, C. Alonso, and F. Guinjoan, "Maximum power point tracking based on slidingmode control for output-series connected converters in photovoltaic systems," *IET Power Electronics*, vol. 7, no. 4, pp. 914-923, April 2014.
- [3] M.A.A. Mohd Zainuri, M.A. Mohd Radzi, A.C. Soh, and N.A. Rahim, "Development of adaptive perturb and observe-fuzzy control maximum power point tracking for photovoltaic boost dc-dc converter," *IET Renewable Power Generation*, vol. 8, no. 2, pp.183-194, March 2014.
- [4] S.K. Kollimalla, M.K. Mishra, "Variable Perturbation Size Adaptive P&O MPPT Algorithm for Sudden Changes in Irradiance," *IEEE Transactions on Sustainable Energy*, vol. 5, no. 3, pp. 718,728, July 2014.
- [5] M.A. Elgendy, B. Zahawi, and D.J. Atkinson, "Assessment of the Incremental Conductance Maximum Power Point Tracking Algorithm," *IEEE Transactions on Sustainable Energy*, vol. 4, no. 1, pp. 108-117, Jan. 2013.
- [6] D.C. Jones, R.W. Erickson, "Probabilistic Analysis of a Generalized Perturb and Observe Algorithm Featuring Robust Operation in the Presence of Power Curve Traps," *IEEE Transactions on Power Electronics*, vol. 28, no. 6, pp. 2912-2926, June 2013.
- [7] D. Sera, L. Mathe, T. Kerekes, S.V. Spataru, and R. Teodorescu, "On the Perturb-and-Observe and Incremental Conductance MPPT Methods for PV Systems," *IEEE Journal of Photovoltaics*, vol. 3, no. 3, pp. 1070-1078, July 2013.
- [8] M.A.G. de Brito, L. Galotto, L.P. Sampaio, G. de Azevedo e Melo, and C.A. Canesin, "Evaluation of the Main MPPT Techniques for Photovoltaic Applications," *IEEE Transactions on Industrial Electronics*, vol. 60, no. 3, pp. 1156-1167, March 2013.
- [9] M.A. Elgendy, B. Zahawi, and D.J. Atkinson, "Assessment of Perturb and Observe MPPT Algorithm Implementation Techniques for PV Pumping Applications," *IEEE Transactions on Sustainable Energy*, vol. 3, no. 1, pp. 21-33, Jan. 2012.
- [10] D.L. King, W.E. Boyson, and J.A. Kratochavil, "Photovoltaic array performance model", Sandia National Laboratories, Technical Report, SAND2004.
- [11] M.M. El-Wakil, "Power plant technology", McGraw-Hill, New York, 1984.
- [12] T. Ishikawa, "Grid-connected photovoltaic power systems: survey of inverter and related protection equipments", Report IEA (International Energy Agency) PVPS, 2002. <http://www.iea-pvps.org>
- [13] Solarbuzz, "solar Cell Technologies", 2010. <http://www.solarbuzz.com/technologies.htm>
- [14] R. Menzies, "Designing a solar power systems", Proc. Solar Photovoltaic Energy Workshop'98, Monash University, Caulfield, 1998.
- [15] Bright Green Energy, "What type of solar PV panels are available?", 2009. http://www.wirefreedirect.com/Types_Solar_PV_Panels.asp

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